in your working memory. This matters, because anything that occupies your working memory reduces your ability to think. You should treat "System 1" and "System 2" as nicknames, like Bob and Joe, identifying characters that you will get to know over the course of this book. The fictitious systems make it easier for me to think about judgment and choice, and will make it easier for you to understand what I say.

### SPEAKING OF SYSTEM 1 AND SYSTEM 2

"He had an impression, but some of his impressions are illusions."

"This was a pure System 1 response. She reacted to the threat before she recognized it."

"This is your System 1 talking. Slow down and let your System 2 take control."

# 2

# ATTENTION AND EFFORT

In the unlikely event of this book being made into a film, System 2 would be a supporting character who believes herself to be the hero. The defining feature of System 2, in this story, is that its operations are effortful, and one of its main characteristics is laziness, a reluctance to invest more effort than is strictly necessary. As a consequence, the thoughts and actions that System 2 believes it has chosen are often guided by the figure at the center of the story, System 1. However, there are vital tasks that only System 2 can perform because they require effort and acts of self-control in which the intuitions and impulses of System 1 are overcome.

## MENTAL EFFORT

If you wish to experience your System 2 working at full tilt, the following exercise will do; it should bring you to the limits of your cognitive abilities within 5 seconds. To start, make up several strings of 4 digits, all different, and write each string on an index card. Place a blank card on top of the deck. The task that you will perform is called Add-1. Here is how it goes:

> Start beating a steady rhythm (or better yet, set a metronome at 1/sec). Remove the blank card and read the four digits aloud. Wait for two beats, then report a string in which each of the original digits is incremented by 1. If the digits on the card are 5294, the correct response is 6305. Keeping the rhythm is important.

Few people can cope with more than four digits in the Add-1 task, but if you want a harder challenge, please try Add-3.

If you would like to know what your body is doing while your mind is hard at work, set up two piles of books on a sturdy table, place a video camera on one and lean your chin on the other, get the video going, and stare at the camera lens while you work on Add-1 or Add-3 exercises. Later, you will find in the changing size of your pupils a faithful record of how hard you worked.

I have a long personal history with the Add-1 task. Early in my career I spent a year at the University of Michigan, as a visitor in a laboratory that studied hypnosis. Casting about for a useful topic of research, I found an article in *Scientific American* in which the psychologist Eckhard Hess described the pupil of the eye as a window to the soul. I reread it recently and again found it inspiring. It begins with Hess reporting that his wife had noticed his pupils widening as he watched beautiful nature pictures, and it ends with two striking pictures of the same good-looking woman, who somehow appears much more attractive in one than in the other. There is only one difference: the pupils of the eyes appear dilated in the attractive picture and constricted in the other. Hess also wrote of belladonna, a pupil-dilating substance that was used as a cosmetic, and of bazaar shoppers who wear dark glasses in order to hide their level of interest from merchants.

One of Hess's findings especially captured my attention. He had noticed that the pupils are sensitive indicators of mental effort—they dilate substantially when people multiply two-digit numbers, and they dilate more if the problems are hard than if they are easy. His observations indicated that the response to mental effort is distinct from emotional arousal. Hess's work did not have much to do with hypnosis, but I concluded that the idea of a visible indication of mental effort had promise as a research topic. A graduate student in the lab, Jackson Beatty, shared my enthusiasm and we got to work.

Beatty and I developed a setup similar to an optician's examination room, in which the experimental participant leaned her head on a chin-and-forehead rest and stared at a camera while listening to prerecorded information and answering questions on the recorded beats of a metronome. The beats triggered an infrared flash every second, causing a picture to be taken. At the end of each experimental session, we would rush to have the film developed, project the images of the pupil on a screen, and go to work with a ruler. The method was a perfect fit for young and impatient



researchers: we knew our results almost immediately, and they always told a clear story.

Beatty and I focused on paced tasks, such as Add-1, in which we knew precisely what was on the subject's mind at any time. We recorded strings of digits on beats of the metronome and instructed the subject to repeat or transform the digits one by one, maintaining the same rhythm. We soon discovered that the size of the pupil varied second by second, reflecting the changing demands of the task. The shape of the response was an inverted V. As you experienced it if you tried Add-1 or Add-3, effort builds up with every added digit that you hear, reaches an almost intolerable peak as you rush to produce a transformed string during and immediately after the pause, and relaxes gradually as you "unload" your short-term memory. The pupil data corresponded precisely to subjective experience: longer strings reliably caused larger dilations, the transformation task compounded the effort, and the peak of pupil size coincided with maximum effort. Add-1 with four digits caused a larger dilation than the task of holding seven digits for immediate recall. Add-3, which is much more difficult, is the most demanding that I ever observed. In the first 5 seconds, the pupil dilates by about 50% of its original area and heart rate increases by about 7 beats per minute. This is as hard as people can work—they give up if more is asked of them. When we exposed our subjects to more digits than they could remember, their pupils stopped dilating or actually shrank.

We worked for some months in a spacious basement suite in which we had set up a closed-circuit system that projected an image of the subject's pupil on a screen in the corridor; we also could hear what was happening in the laboratory. The diameter of the projected pupil was about a foot; watching it dilate and contract when the participant was at work was a fascinating sight, quite an attraction for visitors in our lab. We amused ourselves and impressed our guests by our ability to divine when the participant gave up on a task. During a mental multiplication, the pupil normally dilated to a large size within a few seconds and stayed large as long as the individual kept working on the problem; it contracted immediately when she found a solution or gave up. As we watched from the corridor, we would sometimes surprise both the owner of the pupil and our guests by asking, "Why did you stop working just now?" The answer from inside the lab was often, "How did you know?" to which we would reply, "We have a window to your soul."

The casual observations we made from the corridor were sometimes as informative as the formal experiments. I made a significant discovery as I

was idly watching a woman's pupil during a break between two tasks. She had kept her position on the chin rest, so I could see the image of her eye while she engaged in routine conversation with the experimenter. I was surprised to see that the pupil remained small and did not noticeably dilate as she talked and listened. Unlike the tasks that we were studying, the mundane conversation apparently demanded little or no effort—no more than retaining two or three digits. This was a eureka moment: I realized that the tasks we had chosen for study were exceptionally effortful. An image came to mind: mental life—today I would speak of the life of System 2—is normally conducted at the pace of a comfortable walk, sometimes interrupted by episodes of jogging and on rare occasions by a frantic sprint. The Add-1 and Add-3 exercises are sprints, and casual chatting is a stroll.

We found that people, when engaged in a mental sprint, may become effectively blind. The authors of *The Invisible Gorilla* had made the gorilla "invisible" by keeping the observers intensely busy counting passes. We reported a rather less dramatic example of blindness during Add-1. Our subjects were exposed to a series of rapidly flashing letters while they worked. They were told to give the task complete priority, but they were also asked to report, at the end of the digit task, whether the letter *K* had appeared at any time during the trial. The main finding was that the ability to detect and report the target letter changed in the course of the 10 seconds of the exercise. The observers almost never missed a *K* that was shown at the beginning or near the end of the Add-1 task but they missed the target almost half the time when mental effort was at its peak, although we had pictures of their wide-open eye staring straight at it. Failures of detection followed the same inverted-V pattern as the dilating pupil. The similarity was reassuring: the pupil was a good measure of the physical arousal that accompanies mental effort, and we could go ahead and use it to understand how the mind works.

Much like the electricity meter outside your house or apartment, the pupils offer an index of the current rate at which mental energy is used. The analogy goes deep. Your use of electricity depends on what you choose to do, whether to light a room or toast a piece of bread. When you turn on a bulb or a toaster, it draws the energy it needs but no more. Similarly, we decide what to do, but we have limited control over the effort of doing it. Suppose you are shown four digits, say, 9462, and told that your life depends on holding them in memory for 10 seconds. However much you want to live, you cannot exert as much effort in this task as you would be forced to invest to complete an Add-3 transformation on the same digits.



System 2 and the electrical circuits in your home both have limited capacity, but they respond differently to threatened overload. A breaker trips when the demand for current is excessive, causing all devices on that circuit to lose power at once. In contrast, the response to mental overload is selective and precise: System 2 protects the most important activity, so it receives the attention it needs; "spare capacity" is allocated second by second to other tasks. In our version of the gorilla experiment, we instructed the participants to assign priority to the digit task. We know that they followed that instruction, because the timing of the visual target had no effect on the main task. If the critical letter was presented at a time of high demand, the subjects simply did not see it. When the transformation task was less demanding, detection performance was better.

The sophisticated allocation of attention has been honed by a long evolutionary history. Orienting and responding quickly to the gravest threats or most promising opportunities improved the chance of survival, and this capability is certainly not restricted to humans. Even in modern humans, System 1 takes over in emergencies and assigns total priority to self-protective actions. Imagine yourself at the wheel of a car that unexpectedly skids on a large oil slick. You will find that you have responded to the threat before you became fully conscious of it.

Beatty and I worked together for only a year, but our collaboration had a large effect on our subsequent careers. He eventually became the leading authority on "cognitive pupillometry," and I wrote a book titled *Attention and Effort*, which was based in large part on what we learned together and on follow-up research I did at Harvard the following year. We learned a great deal about the working mind—which I now think of as System 2—from measuring pupils in a wide variety of tasks.

As you become skilled in a task, its demand for energy diminishes. Studies of the brain have shown that the pattern of activity associated with an action changes as skill increases, with fewer brain regions involved. Talent has similar effects. Highly intelligent individuals need less effort to solve the same problems, as indicated by both pupil size and brain activity. A general "law of least effort" applies to cognitive as well as physical exertion. The law asserts that if there are several ways of achieving the same goal, people will eventually gravitate to the least demanding course of action. In the economy of action, effort is a cost, and the acquisition of skill is driven by the balance of benefits and costs. Laziness is built deep into our nature.

The tasks that we studied varied considerably in their effects on the pupil. At baseline, our subjects were awake, aware, and ready to engage in a

task—probably at a higher level of arousal and cognitive readiness than usual. Holding one or two digits in memory or learning to associate a word with a digit (3 = door) produced reliable effects on momentary arousal above that baseline, but the effects were minuscule, only 5% of the increase in pupil diameter associated with Add-3. A task that required discriminating between the pitch of two tones yielded significantly larger dilations. Recent research has shown that inhibiting the tendency to read distracting words (as in figure 2 of the preceding chapter) also induces moderate effort. Tests of short-term memory for six or seven digits were more effortful. As you can experience, the request to retrieve and say aloud your phone number or your spouse's birthday also requires a brief but significant effort, because the entire string must be held in memory as a response is organized. Mental multiplication of two-digit numbers and the Add-3 task are near the limit of what most people can do.

What makes some cognitive operations more demanding and effortful than others? What outcomes must we purchase in the currency of attention? What can System 2 do that System 1 cannot? We now have tentative answers to these questions.

Effort is required to maintain simultaneously in memory several ideas that require separate actions, or that need to be combined according to a rule—rehearsing your shopping list as you enter the supermarket, choosing between the fish and the veal at a restaurant, or combining a surprising result from a survey with the information that the sample was small, for example. System 2 is the only one that can follow rules, compare objects on several attributes, and make deliberate choices between options. The automatic System 1 does not have these capabilities. System 1 detects simple relations ("they are all alike," "the son is much taller than the father") and excels at integrating information about one thing, but it does not deal with multiple distinct topics at once, nor is it adept at using purely statistical information. System 1 will detect that a person described as "a meek and tidy soul, with a need for order and structure, and a passion for detail" resembles a caricature librarian, but combining this intuition with knowledge about the small number of librarians is a task that only System 2 can perform—if System 2 knows how to do so, which is true of few people.

A crucial capability of System 2 is the adoption of "task sets": it can program memory to obey an instruction that overrides habitual responses. Consider the following: Count all occurrences of the letter *f* in this page. This is not a task you have ever performed before and it will not come naturally to you, but your System 2 can take it on. It will be effortful to set



yourself up for this exercise, and effortful to carry it out, though you will surely improve with practice. Psychologists speak of "executive control" to describe the adoption and termination of task sets, and neuroscientists have identified the main regions of the brain that serve the executive function. One of these regions is involved whenever a conflict must be resolved. Another is the prefrontal area of the brain, a region that is substantially more developed in humans than in other primates, and is involved in operations that we associate with intelligence.

Now suppose that at the end of the page you get another instruction: count all the commas in the next page. This will be harder, because you will have to overcome the newly acquired tendency to focus attention on the letter *f*. One of the significant discoveries of cognitive psychologists in recent decades is that switching from one task to another is effortful, especially under time pressure. The need for rapid switching is one of the reasons that Add-3 and mental multiplication are so difficult. To perform the Add-3 task, you must hold several digits in your working memory at the same time, associating each with a particular operation: some digits are in the queue to be transformed, one is in the process of transformation, and others, already transformed, are retained for reporting. Modern tests of working memory require the individual to switch repeatedly between two demanding tasks, retaining the results of one operation while performing the other. People who do well on these tests tend to do well on tests of general intelligence. However, the ability to control attention is not simply a measure of intelligence; measures of efficiency in the control of attention predict performance of air traffic controllers and of Israeli Air Force pilots beyond the effects of intelligence.

Time pressure is another driver of effort. As you carried out the Add-3 exercise, the rush was imposed in part by the metronome and in part by the load on memory. Like a juggler with several balls in the air, you cannot afford to slow down; the rate at which material decays in memory forces the pace, driving you to refresh and rehearse information before it is lost. Any task that requires you to keep several ideas in mind at the same time has the same hurried character. Unless you have the good fortune of a capacious working memory, you may be forced to work uncomfortably hard. The most effortful forms of slow thinking are those that require you to think fast.

You surely observed as you performed Add-3 how unusual it is for your mind to work so hard. Even if you think for a living, few of the mental tasks in which you engage in the course of a working day are as demanding as

Add-3, or even as demanding as storing six digits for immediate recall. We normally avoid mental overload by dividing our tasks into multiple easy steps, committing intermediate results to long-term memory or to paper rather than to an easily overloaded working memory. We cover long distances by taking our time and conduct our mental lives by the law of least effort.

### SPEAKING OF ATTENTION AND EFFORT

"I won't try to solve this while driving. This is a pupil-dilating task. It requires mental effort!"

"The law of least effort is operating here. He will think as little as possible."

"She did not forget about the meeting. She was completely focused on something else when the meeting was set and she just didn't hear you."

"What came quickly to my mind was an intuition from System 1. I'll have to start over and search my memory deliberately."

# 3

# THE LAZY CONTROLLER

I spend a few months each year in Berkeley, and one of my great pleasures there is a daily four-mile walk on a marked path in the hills, with a fine view of San Francisco Bay. I usually keep track of my time and have learned a fair amount about effort from doing so. I have found a speed, about 17 minutes for a mile, which I experience as a stroll. I certainly exert physical effort and burn more calories at that speed than if I sat in a recliner, but I experience no strain, no conflict, and no need to push myself. I am also able to think and work while walking at that rate. Indeed, I suspect that the mild physical arousal of the walk may spill over into greater mental alertness.

System 2 also has a natural speed. You expend some mental energy in random thoughts and in monitoring what goes on around you even when your mind does nothing in particular, but there is little strain. Unless you are in a situation that makes you unusually wary or self-conscious, monitoring what happens in the environment or inside your head demands little effort. You make many small decisions as you drive your car, absorb some information as you read the newspaper, and conduct routine exchanges of pleasantries with a spouse or a colleague, all with little effort and no strain. Just like a stroll.

It is normally easy and actually quite pleasant to walk and think at the same time, but at the extremes these activities appear to compete for the limited resources of System 2. You can confirm this claim by a simple experiment. While walking comfortably with a friend, ask him to compute $23 \times 78$ in his head, and to do so immediately. He will almost certainly stop

in his tracks. My experience is that I can think while strolling but cannot engage in mental work that imposes a heavy load on short-term memory. If I must construct an intricate argument under time pressure, I would rather be still, and I would prefer sitting to standing. Of course, not all slow thinking requires that form of intense concentration and effortful computation—I did the best thinking of my life on leisurely walks with Amos.

Accelerating beyond my strolling speed completely changes the experience of walking, because the transition to a faster walk brings about a sharp deterioration in my ability to think coherently. As I speed up, my attention is drawn with increasing frequency to the experience of walking and to the deliberate maintenance of the faster pace. My ability to bring a train of thought to a conclusion is impaired accordingly. At the highest speed I can sustain on the hills, about 14 minutes for a mile, I do not even try to think of anything else. In addition to the physical effort of moving my body rapidly along the path, a mental effort of self-control is needed to resist the urge to slow down. Self-control and deliberate thought apparently draw on the same limited budget of effort.

For most of us, most of the time, the maintenance of a coherent train of thought and the occasional engagement in effortful thinking also require self-control. Although I have not conducted a systematic survey, I suspect that frequent switching of tasks and speeded-up mental work are not intrinsically pleasurable, and that people avoid them when possible. This is how the law of least effort comes to be a law. Even in the absence of time pressure, maintaining a coherent train of thought requires discipline. An observer of the number of times I look at e-mail or investigate the refrigerator during an hour of writing could reasonably infer an urge to escape and conclude that keeping at it requires more self-control than I can readily muster.

Fortunately, cognitive work is not always aversive, and people sometimes expend considerable effort for long periods of time without having to exert willpower. The psychologist Mihaly Csikszentmihalyi (pronounced six-cent-mihaly) has done more than anyone else to study this state of effortless attending, and the name he proposed for it, *flow*, has become part of the language. People who experience flow describe it as "a state of effortless concentration so deep that they lose their sense of time, of themselves, of their problems," and their descriptions of the joy of that state are so compelling that Csikszentmihalyi has called it an "optimal experience." Many activities can induce a sense of flow, from painting to racing motorcycles—and for some fortunate authors I know, even writing a book is often an optimal experience. Flow neatly separates the two forms of effort: concentration on the task and the deliberate control of attention. Riding a motorcycle at 150 miles an hour and playing a competitive game of chess are certainly very effortful. In a state of flow, however, maintaining focused attention on these absorbing activities requires no exertion of self-control, thereby freeing resources to be directed to the task at hand.



## THE BUSY AND DEPLETED SYSTEM 2

It is now a well-established proposition that both self-control and cognitive effort are forms of mental work. Several psychological studies have shown that people who are simultaneously challenged by a demanding cognitive task and by a temptation are more likely to yield to the temptation. Imagine that you are asked to retain a list of seven digits for a minute or two. You are told that remembering the digits is your top priority. While your attention is focused on the digits, you are offered a choice between two desserts: a sinful chocolate cake and a virtuous fruit salad. The evidence suggests that you would be more likely to select the tempting chocolate cake when your mind is loaded with digits. System 1 has more influence on behavior when System 2 is busy, and it has a sweet tooth.

People who are *cognitively busy* are also more likely to make selfish choices, use sexist language, and make superficial judgments in social situations. Memorizing and repeating digits loosens the hold of System 2 on behavior, but of course cognitive load is not the only cause of weakened self-control. A few drinks have the same effect, as does a sleepless night. The self-control of morning people is impaired at night; the reverse is true of night people. Too much concern about how well one is doing in a task sometimes disrupts performance by loading short-term memory with pointless anxious thoughts. The conclusion is straightforward: self-control requires attention and effort. Another way of saying this is that controlling thoughts and behaviors is one of the tasks that System 2 performs.

A series of surprising experiments by the psychologist Roy Baumeister and his colleagues has shown conclusively that all variants of voluntary effort—cognitive, emotional, or physical—draw at least partly on a shared pool of mental energy. Their experiments involve successive rather than simultaneous tasks.

Baumeister's group has repeatedly found that an effort of will or self-control is tiring; if you have had to force yourself to do something, you are less willing or less able to exert self-control when the next challenge comes

around. The phenomenon has been named *ego depletion*. In a typical demonstration, participants who are instructed to stifle their emotional reaction to an emotionally charged film will later perform poorly on a test of physical stamina—how long they can maintain a strong grip on a dynamometer in spite of increasing discomfort. The emotional effort in the first phase of the experiment reduces the ability to withstand the pain of sustained muscle contraction, and ego-depleted people therefore succumb more quickly to the urge to quit. In another experiment, people are first depleted by a task in which they eat virtuous foods such as radishes and celery while resisting the temptation to indulge in chocolate and rich cookies. Later, these people will give up earlier than normal when faced with a difficult cognitive task.

The list of situations and tasks that are now known to deplete self-control is long and varied. All involve conflict and the need to suppress a natural tendency. They include:

avoiding the thought of white bears
inhibiting the emotional response to a stirring film
making a series of choices that involve conflict
trying to impress others
responding kindly to a partner's bad behavior
interacting with a person of a different race (for prejudiced individuals)

The list of indications of depletion is also highly diverse:

deviating from one's diet
overspending on impulsive purchases
reacting aggressively to provocation
persisting less time in a handgrip task
performing poorly in cognitive tasks and logical decision making

The evidence is persuasive: activities that impose high demands on System 2 require self-control, and the exertion of self-control is depleting and unpleasant. Unlike cognitive load, ego depletion is at least in part a loss of motivation. After exerting self-control in one task, you do not feel like making an effort in another, although you could do it if you really had to. In several experiments, people were able to resist the effects of ego depletion when given a strong incentive to do so. In contrast, increasing effort is not an option when you must keep six digits in short-term memory while per-



forming a task. Ego depletion is not the same mental state as cognitive busyness.

The most surprising discovery made by Baumeister's group shows, as he puts it, that the idea of mental energy is more than a mere metaphor. The nervous system consumes more glucose than most other parts of the body, and effortful mental activity appears to be especially expensive in the currency of glucose. When you are actively involved in difficult cognitive reasoning or engaged in a task that requires self-control, your blood glucose level drops. The effect is analogous to a runner who draws down glucose stored in her muscles during a sprint. The bold implication of this idea is that the effects of ego depletion could be undone by ingesting glucose, and Baumeister and his colleagues have confirmed this hypothesis in several experiments.

Volunteers in one of their studies watched a short silent film of a woman being interviewed and were asked to interpret her body language. While they were performing the task, a series of words crossed the screen in slow succession. The participants were specifically instructed to ignore the words, and if they found their attention drawn away they had to refocus their concentration on the woman's behavior. This act of self-control was known to cause ego depletion. All the volunteers drank some lemonade before participating in a second task. The lemonade was sweetened with glucose for half of them and with Splenda for the others. Then all participants were given a task in which they needed to overcome an intuitive response to get the correct answer. Intuitive errors are normally much more frequent among ego-depleted people, and the drinkers of Splenda showed the expected depletion effect. On the other hand, the glucose drinkers were not depleted. Restoring the level of available sugar in the brain had prevented the deterioration of performance. It will take some time and much further research to establish whether the tasks that cause glucose-depletion also cause the momentary arousal that is reflected in increases of pupil size and heart rate.

A disturbing demonstration of depletion effects in judgment was recently reported in the *Proceedings of the National Academy of Sciences*. The unwitting participants in the study were eight parole judges in Israel. They spend entire days reviewing applications for parole. The cases are presented in random order, and the judges spend little time on each one, an average of 6 minutes. (The default decision is denial of parole; only 35% of requests are approved. The exact time of each decision is recorded, and the times of the judges' three food breaks—morning break, lunch, and afternoon break—during the day are recorded as well.) The authors of the study plotted the

proportion of approved requests against the time since the last food break. The proportion spikes after each meal, when about 65% of requests are granted. During the two hours or so until the judges' next feeding, the approval rate drops steadily, to about zero just before the meal. As you might expect, this is an unwelcome result and the authors carefully checked many alternative explanations. The best possible account of the data provides bad news: tired and hungry judges tend to fall back on the easier default position of denying requests for parole. Both fatigue and hunger probably play a role.

## THE LAZY SYSTEM 2

One of the main functions of System 2 is to monitor and control thoughts and actions "suggested" by System 1, allowing some to be expressed directly in behavior and suppressing or modifying others.

For an example, here is a simple puzzle. Do not try to solve it but listen to your intuition:

> A bat and ball cost $1.10.
> The bat costs one dollar more than the ball.
> How much does the ball cost?

A number came to your mind. The number, of course, is 10: 10¢. The distinctive mark of this easy puzzle is that it evokes an answer that is intuitive, appealing, and wrong. Do the math, and you will see. If the ball costs 10¢, then the total cost will be $1.20 (10¢ for the ball and $1.10 for the bat), not $1.10. The correct answer is 5¢. It is safe to assume that the intuitive answer also came to the mind of those who ended up with the correct number—they somehow managed to resist the intuition.

Shane Frederick and I worked together on a theory of judgment based on two systems, and he used the bat-and-ball puzzle to study a central question: How closely does System 2 monitor the suggestions of System 1? His reasoning was that we know a significant fact about anyone who says that the ball costs 10¢: that person did not actively check whether the answer was correct, and her System 2 endorsed an intuitive answer that it could have rejected with a small investment of effort. Furthermore, we also know that the people who give the intuitive answer have missed an obvious social cue; they should have wondered why anyone would include in a questionnaire a puzzle with such an obvious answer. A failure to check is remarkable



because the cost of checking is so low: a few seconds of mental work (the problem is moderately difficult), with slightly tensed muscles and dilated pupils, could avoid an embarrassing mistake. People who say 10¢ appear to be ardent followers of the law of least effort. People who avoid that answer appear to have more active minds.

Many thousands of university students have answered the bat-and-ball puzzle, and the results are shocking. More than 50% of students at Harvard, MIT, and Princeton gave the intuitive—incorrect—answer. At less selective universities, the rate of demonstrable failure to check was in excess of 80%. The bat-and-ball problem is our first encounter with an observation that will be a recurrent theme of this book: many people are overconfident, prone to place too much faith in their intuitions. They apparently find cognitive effort at least mildly unpleasant and avoid it as much as possible.

Now I will show you a logical argument—two premises and a conclusion. Try to determine, as quickly as you can, if the argument is logically valid. Does the conclusion follow from the premises?

> All roses are flowers.
> Some flowers fade quickly.
> Therefore some roses fade quickly.

A large majority of college students endorse this syllogism as valid. In fact the argument is flawed, because it is possible that there are no roses among the flowers that fade quickly. Just as in the bat-and-ball problem, a plausible answer comes to mind immediately. Overriding it requires hard work—the insistent idea that "it's true, it's true!" makes it difficult to check the logic, and most people do not take the trouble to think through the problem.

This experiment has discouraging implications for reasoning in everyday life. It suggests that when people believe a conclusion is true, they are also very likely to believe arguments that appear to support it, even when these arguments are unsound. If System 1 is involved, the conclusion comes first and the arguments follow.

Next, consider the following question and answer it quickly before reading on:

> How many murders occur in the state of Michigan in one year?

The question, which was also devised by Shane Frederick, is again a challenge to System 2. The "trick" is whether the respondent will remember that

Detroit, a high-crime city, is in Michigan. College students in the United States know this fact and will correctly identify Detroit as the largest city in Michigan. But knowledge of a fact is not all-or-none. Facts that we know do not always come to mind when we need them. People who remember that Detroit is in Michigan give higher estimates of the murder rate in the state than people who do not, but a majority of Frederick's respondents did not think of the city when questioned about the state. Indeed, the average guess by people who were asked about Michigan is *lower* than the guesses of a similar group who were asked about the murder rate in Detroit.

Blame for a failure to think of Detroit can be laid on both System 1 and System 2. Whether the city comes to mind when the state is mentioned depends in part on the automatic function of memory. People differ in this respect. The representation of the state of Michigan is very detailed in some people's minds: residents of the state are more likely to retrieve many facts about it than people who live elsewhere; geography buffs will retrieve more than others who specialize in baseball statistics; more intelligent individuals are more likely than others to have rich representations of most things. Intelligence is not only the ability to reason; it is also the ability to find relevant material in memory and to deploy attention when needed. Memory function is an attribute of System 1. However, everyone has the option of slowing down to conduct an active search of memory for all possibly relevant facts—just as they could slow down to check the intuitive answer in the bat-and-ball problem. The extent of deliberate checking and search is a characteristic of System 2, which varies among individuals.

The bat-and-ball problem, the flowers syllogism, and the Michigan/Detroit problem have something in common. Failing these minitests appears to be, at least to some extent, a matter of insufficient motivation, not trying hard enough. Anyone who can be admitted to a good university is certainly able to reason through the first two questions and to reflect about Michigan long enough to remember the major city in that state and its crime problem. These students can solve much more difficult problems when they are not tempted to accept a superficially plausible answer that comes readily to mind. The ease with which they are satisfied enough to stop thinking is rather troubling. "Lazy" is a harsh judgment about the self-monitoring of these young people and their System 2, but it does not seem to be unfair. Those who avoid the sin of intellectual sloth could be called "engaged." They are more alert, more intellectually active, less willing to be satisfied with superficially attractive answers, more skeptical about their intuitions. The psychologist Keith Stanovich would call them more rational.



## INTELLIGENCE, CONTROL, RATIONALITY

Researchers have applied diverse methods to examine the connection between thinking and self-control. Some have addressed it by asking the correlation question: If people were ranked by their self-control and by their cognitive aptitude, would individuals have similar positions in the two rankings?

In one of the most famous experiments in the history of psychology, Walter Mischel and his students exposed four-year-old children to a cruel dilemma. They were given a choice between a small reward (one Oreo), which they could have at any time, or a larger reward (two cookies) for which they had to wait 15 minutes under difficult conditions. They were to remain alone in a room, facing a desk with two objects: a single cookie and a bell that the child could ring at any time to call in the experimenter and receive the one cookie. As the experiment was described: "There were no toys, books, pictures, or other potentially distracting items in the room. The experimenter left the room and did not return until 15 min had passed or the child had rung the bell, eaten the rewards, stood up, or shown any signs of distress."

The children were watched through a one-way mirror, and the film that shows their behavior during the waiting time always has the audience roaring in laughter. About half the children managed the feat of waiting for 15 minutes, mainly by keeping their attention away from the tempting reward. Ten or fifteen years later, a large gap had opened between those who had resisted temptation and those who had not. The resisters had higher measures of executive control in cognitive tasks, and especially the ability to reallocate their attention effectively. As young adults, they were less likely to take drugs. A significant difference in intellectual aptitude emerged: the children who had shown more self-control as four-year-olds had substantially higher scores on tests of intelligence.

A team of researchers at the University of Oregon explored the link between cognitive control and intelligence in several ways, including an attempt to raise intelligence by improving the control of attention. During five 40-minute sessions, they exposed children aged four to six to various computer games especially designed to demand attention and control. In one of the exercises, the children used a joystick to track a cartoon cat and move it to a grassy area while avoiding a muddy area. The grassy areas gradually shrank and the muddy area expanded, requiring progressively more precise control. The testers found that training attention not only improved

executive control; scores on nonverbal tests of intelligence also improved and the improvement was maintained for several months. Other research by the same group identified specific genes that are involved in the control of attention, showed that parenting techniques also affected this ability, and demonstrated a close connection between the children's ability to control their attention and their ability to control their emotions.

Shane Frederick constructed a Cognitive Reflection Test, which consists of the bat-and-ball problem and two other questions, chosen because they also invite an intuitive answer that is both compelling and wrong (the questions are shown in chapter 5). He went on to study the characteristics of students who score very low on this test—the supervisory function of System 2 is weak in these people—and found that they are prone to answer questions with the first idea that comes to mind and unwilling to invest the effort needed to check their intuitions. Individuals who uncritically follow their intuitions about puzzles are also prone to accept other suggestions from System 1. In particular, they are impulsive, impatient, and keen to receive immediate gratification. For example, 63% of the intuitive respondents say they would prefer to get $3,400 this month rather than $3,800 next month. Only 37% of those who solve all three puzzles correctly have the same shortsighted preference for receiving a smaller amount immediately. When asked how much they will pay to get overnight delivery of a book they have ordered, the low scorers on the Cognitive Reflection Test are willing to pay twice as much as the high scorers. Frederick's findings suggest that the characters of our psychodrama have different "personalities." System 1 is impulsive and intuitive; System 2 is capable of reasoning, and it is cautious, but at least for some people it is also lazy. We recognize related differences among individuals: some people are more like their System 2; others are closer to their System 1. This simple test has emerged as one of the better predictors of lazy thinking.

Keith Stanovich and his longtime collaborator Richard West originally introduced the terms System 1 and System 2 (they now prefer to speak of Type 1 and Type 2 processes). Stanovich and his colleagues have spent decades studying differences among individuals in the kinds of problems with which this book is concerned. They have asked one basic question in many different ways: What makes some people more susceptible than others to biases of judgment? Stanovich published his conclusions in a book titled *Rationality and the Reflective Mind*, which offers a bold and distinctive approach to the topic of this chapter. He draws a sharp distinction between two parts of System 2—indeed, the distinction is so sharp that he calls them



separate "minds." One of these minds (he calls it algorithmic) deals with slow thinking and demanding computation. Some people are better than others in these tasks of brain power—they are the individuals who excel in intelligence tests and are able to switch from one task to another quickly and efficiently. However, Stanovich argues that high intelligence does not make people immune to biases. Another ability is involved, which he labels rationality. Stanovich's concept of a rational person is similar to what I earlier labeled "engaged." The core of his argument is that *rationality* should be distinguished from *intelligence*. In his view, superficial or "lazy" thinking is a flaw in the reflective mind, a failure of rationality. This is an attractive and thought-provoking idea. In support of it, Stanovich and his colleagues have found that the bat-and-ball question and others like it are somewhat better indicators of our susceptibility to cognitive errors than are conventional measures of intelligence, such as IQ tests. Time will tell whether the distinction between intelligence and rationality can lead to new discoveries.

## SPEAKING OF CONTROL

"She did not have to struggle to stay on task for hours. She was in a state of *flow*."

"His ego was depleted after a long day of meetings. So he just turned to standard operating procedures instead of thinking through the problem."

"He didn't bother to check whether what he said made sense. Does he usually have a lazy System 2 or was he unusually tired?"

"Unfortunately, she tends to say the first thing that comes into her mind. She probably also has trouble delaying gratification. Weak System 2."